



Lecture 1: What is NLP?

CS 584

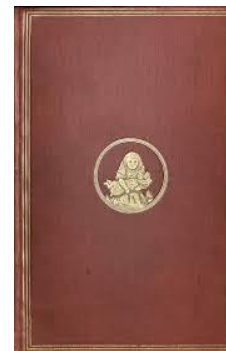
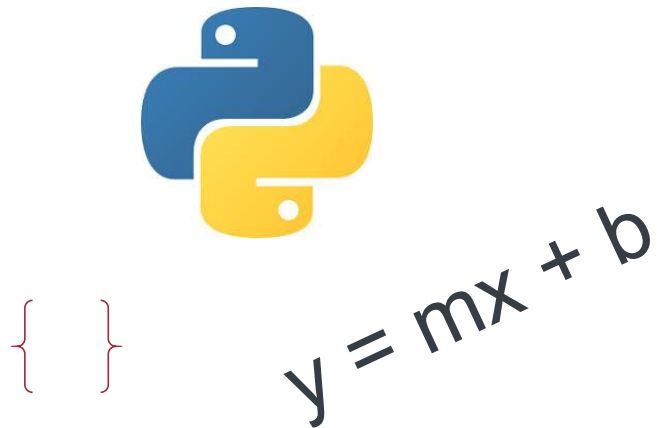
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What is NLP?

Using formal language to process and analyze natural language



“ETA is around 8:20”

What is Language?

- A communication tool

”Doing” NLP

- Using computers, math, and statistics to process natural language

ELIZA: Early NLP via Simple Patterns

User: I need some help, that much seems certain.

ELIZA: WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP

User: Perhaps I could learn to get along with my mother.

ELIZA: TELL ME MORE ABOUT YOUR FAMILY

ELIZA was basically pattern matching on **words** and simple templates.

It had no real understanding, but users still felt understood.

This raises our first question: **what exactly is a “word”?**

Project connection: our models are also driven by **patterns over tokens**, not true understanding.

Units of Language: Symbols, Words, Tokens, etc.

- Alphabet
- Token
- Type
- Word
- Phrase
- Sentence
- Document
- Corpus



Texts, Sentences, and Utterances

Sentence

Written unit, usually grammatically complete

Often ends in ., ?, or !

Utterance

Spoken unit

May be incomplete or messy:

"I do uh main- mainly business data processing"

Document

Collection of sentences or utterances (article, review, chapter, transcript)

Project connection:

Your tokenizer must operate on **text streams**

Your language model will be trained on **documents**, but generate **sentences/utterances**



Are Words Universal?

English: spaces mark word boundaries pretty well.

Chinese:

No spaces between characters

The same sequence can be segmented in different ways (3-word vs 5-word analyses)

Spoken English:

Disfluencies: *uh*, *um*, *main-*

Are *uh* and *um* words? Sometimes yes (for speech recognition), sometimes no.

Takeaway: “word” is a **useful but messy** concept, especially across languages and modalities.

Project connection: this is why modern systems use **subword tokens**, not just words. We will build toward that.



Thank you!